TUE MINH CAO

Mechanistic Interpretability & AI Safety Research · PhD Student, University of Florida

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Summary

Mechanistic interpretability researcher specializing in sparse autoencoders, circuit tracing, activation and attribution patching, and weight-space decomposition. Develops PyTorch-based methods for understanding, evaluating, and auditing large language models, including the mechanisms behind hallucination and misaligned behavior. First-author ICLR 2025 paper, with four further papers under review at NeurIPS, EMNLP, and IEEE T-ITS.

Education

University of Florida *Jan 2026 – Present*
PhD, Computer Science · Advisor: Prof. My T. Thai · GPA 4.0 / 4.0

Hanoi University of Science and Technology *Sep 2021 – Sep 2025*
BSc, Computer Science (Talent Program) · GPA 3.86 / 4.0 · first-author ICLR 2025 paper as an undergraduate

Publications

- **NeurFlow: Interpreting Neural Networks through Critical Neuron Groups and Functional Interactions** *(ICLR 2025)* — **Cao, T. M.**, Xuan, N. H., Nguyen, L. P., Pham, H. H., Thai, M. T.
  Replaces the neuron with the neuron group as the unit of circuit analysis. Patching a group of only 4 neurons removes a concept from the model, while patching individual neurons does not. Surfaced background-bias circuits in a vision model that existing methods missed.
- **Semantic Optimal Transport for Sparse Autoencoder Feature Matching and Circuit Compression** *(under review, NeurIPS 2026)* — **Cao, T. M.**, Do, N., Thai, M. T.
  Matches semantically equivalent SAE features across layers via optimal transport, enabling automatic supernode compression of feature circuits under proven stability and matching-optimality guarantees. Compressed open-source Neuronpedia circuits to 10% of their original size and improved matching performance by 20–30% over prior state of the art.
- **Tree SAE: Learning Hierarchical Feature Structures in Sparse Autoencoders** *(under review, EMNLP 2026)* — **Cao, T. M.**, Xuan, N. H., Alharbi, R., Nguyen, L. P., Thai, M. T.
  Shows that parent–child pairs in existing hierarchical SAEs do not share semantics, and introduces a reconstruction-conditioned tree SAE learning genuinely hierarchical features — over 40% more coherent than prior state of the art at matched absorption and feature-splitting scores.
- **ReSS: Learning Reasoning Models for Tabular Data Prediction via Symbolic Scaffold** *(under review, NeurIPS 2026)* — Yi, C., Li, G., Xiong, Z., **Cao, T. M.**, Gong, Y., Thai, M. T., Yang, T.
  Decision-tree-scaffolded reasoning traces for tabular prediction; contributed the faithfulness and causal-sufficiency metrics. Improved accuracy by up to 10% on medical and financial benchmarks while producing faithful, interpretable reasoning.
- **Sphinx: First Explain, Then Explore** *(under review, IEEE T-ITS)* — Do, N., **Cao, T. M.**, Do, T. V., Hajdu, A., Berczes, T., Thai, M. T.
  Explains the failure modes of autonomous driving models — indecisiveness, incorrect visual grounding, and inconsistency between frames — then automatically generates critical scenarios from those failure modes to improve the model, reaching a driving success rate above 80% across multiple models (>30% over prior state of the art).

Research Experience

Graduate Researcher — Mechanistic Interpretability *Jan 2026 – Present*
Adaptive Learning and Optimization Lab, University of Florida · Advisor: Prof. My T. Thai

- Lead 3 concurrent interpretability projects and mentor 5 undergraduate researchers across UF, HUST, and NTU; submitted 3 papers to top-tier venues within the first semester of the PhD.

Interpretable Parameter Decomposition via Activation-Space Supervision *Project lead · Jul 2026 – Present*

- Defined the project direction: use the activation space to decompose weight space into interpretable components.
- Proposed jointly training parameter decomposition with activation-space decomposition using sparse autoencoders, then validated the approach with SAEBench and the interpretability Intruder score.
- Raised the Intruder score of parameter decomposition from 0.2 (low interpretability) to 0.75 (high interpretability) on a large model, over existing methods.

Misalignment Localization in Finetuned Weights *Project lead · Jul 2026 – Present*

- Defined the project direction: locate, inside finetuned weights, the mechanism responsible for emergently misaligned behavior.
- Proposed aggregating gradients over behaviorally similar examples to suppress per-example gradient noise, then decomposing the aggregated gradient with a sparse autoencoder into sparse, interpretable atoms.
- Validated atom-level attributions with circuit tracing, activation patching, and attribution patching; supervise 2 undergraduate researchers.

Self-Reported Misalignment Adapter *Project lead · Jul 2026 – Present*

- Defined the project direction: audit misalignment that lies *outside* the finetuning distribution, where behavioral testing on the finetuning data gives no signal.
- Built an automated auditing agent that elicits out-of-distribution misaligned behavior and assembles it into an OOD misalignment dataset, then trained a model-agnostic LoRA adapter on that dataset so a finetuned model self-reports unforeseen misaligned behavior.

- Structured the evaluation around cross-domain transfer: eliciting confession of medical-domain misalignment from a model finetuned only on misaligned code; supervise 1 undergraduate researcher.

Cross-Layer SAE Feature Matching & Circuit Compression *First author · Under review, NeurIPS 2026*

- Developed a semantic optimal-transport framework that matches semantically equivalent sparse-autoencoder features across layers, and proved stability of the semantic Wasserstein score under noise plus optimality of the matching within a bounded margin.
- Compressed open-source Neuronpedia feature circuits automatically to 10% of their original size, automating a grouping step performed manually in circuit tracing.
- Improved feature-matching performance by 20–30% over prior state of the art; designed all main experiments (LLM-as-judge interpretability evaluation, circuit compression, ground-truth matching).

Faithfulness Evaluation for LLM Reasoning Traces (ReSS) *Contributor · Under review, NeurIPS 2026*

- Designed and implemented the faithfulness and causal-sufficiency metrics for generated reasoning traces, using counterfactual ablation of individual reasoning steps.
- Established that decision-tree-scaffolded traces are causally load-bearing and non-redundant rather than post-hoc rationalization.
- Contributed to a system improving accuracy by up to 10% on medical and financial benchmarks while preserving faithful, interpretable reasoning.

Failure-Mode Diagnosis for a Vision-Language Driving Model (Sphinx) *Research assistant · Under review, IEEE T-ITS*

- Diagnosed failure modes by extracting concepts and attributing them to the input regions carrying each concept, then designed experiments verifying every explanation.
- Identified indecisiveness under urgent conditions, incorrect visual grounding, and inconsistency across frames as the dominant failure modes.
- Built an automated loop turning each failure explanation into targeted test scenarios for retraining, reaching a driving success rate above 80% across multiple models (>30% over prior state of the art).

Researcher — Interpretability (ongoing external collaboration) *Jan 2024 – Present*

AI4LIFE, Hanoi University of Science and Technology · Advisor: Assoc. Prof. Phi Le Nguyen

Weight-Space Source of Object Hallucination in Vision-Language Models *Project lead · Sep 2025 – Present*

- Designed a weight-space decomposition that isolates low-rank regions of weight space encoding the prior knowledge driving object hallucination, and validated the localization with activation and attribution patching.
- Established a negative result: prior-knowledge editing methods (ROME, MEMIT, Nullu) fail to localize this knowledge and induce language collapse.
- Showed that finetuning-based hallucination mitigation introduces new mechanisms rather than removing the source; supervise 2 undergraduate researchers (HUST, NTU).

Tree-Structured Sparse Autoencoders *First author · Under review, EMNLP 2026*

- Identified that parent–child feature pairs in existing hierarchical SAEs do not share semantic meaning, invalidating their claim to hierarchy.
- Designed the first tree-structured sparse autoencoder, using a reconstruction condition to enforce a genuine parent–child relationship between features.
- Produced hierarchical feature pairs over 40% more coherent than prior state of the art at matched absorption and feature-splitting scores; led all phases from formulation to benchmark evaluation.

NeurFlow — Group-Level Circuit Analysis *First author · Accepted, ICLR 2025*

- Formulated circuit analysis over neuron *groups* rather than individual neurons, addressing polysemanticity and low per-neuron contribution.
- Showed that patching a group of only 4 neurons removes a concept from the model, while patching individual neurons does not.
- Designed group-level circuits using gradient attribution to trace how decisions propagate across layers, surfacing background-bias circuits existing methods missed; led all phases through peer review.

Technical Skills

Interpretability: sparse autoencoders, circuit tracing & attribution graphs, activation and attribution patching, causal tracing, weight / parameter decomposition, feature geometry, model auditing & evals, TransformerLens, SAELens

ML engineering: Python, C++, PyTorch, PyTorch Lightning, NumPy, HuggingFace Transformers, LoRA / PEFT finetuning, Weights & Biases, Git, reproducible experiment pipelines, deep learning, machine learning

Foundations: scientific writing, LaTeX

Honors & Awards

- 2nd place, Linear Algebra — National Mathematical Olympiad for University Students, Vietnam Mathematical Society (2022).
- 2nd place, Calculus — National Mathematical Olympiad for University Students, Vietnam Mathematical Society (2022).
- National team selection participant, International Physics Olympiad — Ministry of Education, Vietnam (2021).
- 2nd place, National Physics Olympiad — Ministry of Education, Vietnam (2021).
- 3rd place, National Physics Olympiad — Ministry of Education, Vietnam (2020).
- 1st place, Coastal Provinces & Northern Vietnam High School Competition in Physics (2019).